Solving Linear Systems by Substitution
Answer Key

Algebra 1

Quick Answers

1. $x = 3$, $y = 6$
2. $x = 3$, $y =$ -1
3. $x = 4$, $y = 3$
4. $x = 16$, $y = 5$
5. $x = 3$, $y = 4$
6. $x = 7$, $y = 5$
7. $x = \frac{7}{5}$, $y = \frac{11}{5}$
8. $x = 5$, $y = 2$
9. —
10. $x = 2$, $y = 2$
11. $x = 6$, $y = 17$
12. $x = 8$, $y = 4$

Curriculum

Level: Algebra 1
HSA-REI.C.6 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 12 tagged checks.

1. $y = 2x, \quad 3x + y = 15$
Step 1. The first equation is already solved for y, so replace y in the second: $3x + 2x = 15$.
Step 2. $5x = 15$, so $x = 3$.
Step 3. Back-substitute: $y = 2(3) = 6$. Check in $3x + y$: $9 + 6 = 15$ ✓ $\boxed{(3,\ 6)}$

2. $y = x - 4, \quad 2x + 3y = 3$
Step 1. Substitute $x - 4$ for y: $2x + 3(x - 4) = 3$.
Step 2. Distribute and collect: $2x + 3x - 12 = 3$, so $5x = 15$ and $x = 3$.
Step 3. Back-substitute: $y = 3 - 4 = -1$. Check in $2x + 3y$: $6 - 3 = 3$ ✓ $\boxed{(3,\ -1)}$

3. $2x + y = 11, \quad 3x + 2y = 18$
Step 1. Nothing is isolated yet. The cheapest isolation is y in the first equation (its coefficient is 1): $y = 11 - 2x$.
Step 2. Substitute into the second: $3x + 2(11 - 2x) = 18$, so $3x + 22 - 4x = 18$ and $-x = -4$, giving $x = 4$.
Step 3. Back-substitute: $y = 11 - 8 = 3$. Check in $3x + 2y$: $12 + 6 = 18$ ✓ $\boxed{(4,\ 3)}$

4. $x = 3y + 1, \quad 2x - 5y = 7$
Step 1. Here x is the isolated variable, so substitute for x: $2(3y + 1) - 5y = 7$.
Step 2. $6y + 2 - 5y = 7$, so $y + 2 = 7$ and $y = 5$.
Step 3. Back-substitute: $x = 3(5) + 1 = 16$. Check in $2x - 5y$: $32 - 25 = 7$ ✓ $\boxed{(16,\ 5)}$

5. $x + y = 7, \quad 3x - y = 5$
Step 1. Isolate y in the first equation: $y = 7 - x$.
Step 2. Substitute: $3x - (7 - x) = 5$. The subtraction applies to the whole expression, so $3x - 7 + x = 5$, giving $4x = 12$ and $x = 3$.
Step 3. Back-substitute: $y = 7 - 3 = 4$. Check in $3x - y$: $9 - 4 = 5$ ✓ $\boxed{(3,\ 4)}$

6. Tickets: x adult at 8 dollars, y child at 5 dollars.
Step 1. Counting tickets gives $x + y = 12$; counting dollars gives $8x + 5y = 81$.
Step 2. Isolate in the count equation: $y = 12 - x$, then substitute: $8x + 5(12 - x) = 81$.
Step 3. $8x + 60 - 5x = 81$, so $3x = 21$ and $x = 7$; then $y = 12 - 7 = 5$. Check the money: $56 + 25 = 81$ ✓
The family bought 7 adult tickets and 5 child tickets. $\boxed{(7,\ 5)}$

7. $y = 3x - 2, \quad 2x + y = 5$
Step 1. Substitute for y: $2x + (3x - 2) = 5$.
Step 2. $5x - 2 = 5$, so $5x = 7$ and $x = \frac{7}{5}$. A fraction is a perfectly good solution — do not round it.
Step 3. Back-substitute: $y = 3 \cdot \frac{7}{5} - 2 = \frac{21}{5} - \frac{10}{5} = \frac{11}{5}$. Check in $2x + y$: $\frac{14}{5} + \frac{11}{5} = \frac{25}{5} = 5$ ✓ $\boxed{\left(\frac{7}{5},\ \frac{11}{5}\right)}$

8. $x - 3y = -1, \quad 2x + y = 12$
Step 1. Isolate x in the first equation: $x = 3y - 1$.
Step 2. Substitute: $2(3y - 1) + y = 12$, so $6y - 2 + y = 12$ and $7y = 14$, giving $y = 2$.
Step 3. Back-substitute: $x = 3(2) - 1 = 5$. Check in $2x + y$: $10 + 2 = 12$ ✓ $\boxed{(5,\ 2)}$

9. $y = 2x + 3, \quad 4x - 2y = 1$ — **no solution**
Step 1. Substitute for y: $4x - 2(2x + 3) = 1$.
Step 2. $4x - 4x - 6 = 1$, so $-6 = 1$. Both variables vanished and what is left is *false*.
Step 3. A false statement means no (x, y) can satisfy both equations: the system has **no solution**. Rewriting the second equation as $y = 2x - \frac{1}{2}$ shows why — same slope 2, different intercepts, so the lines are parallel and never meet.
Answer to accept: "No solution; the lines are parallel" with the $-6 = 1$ collapse shown as evidence. $\boxed{\text{no solution}}$

10. $y = 3x - 4, \quad 6x - 2y = 8$ — **infinitely many**
Step 1. Substitute for y: $6x - 2(3x - 4) = 8$.
Step 2. $6x - 6x + 8 = 8$, so $8 = 8$. The variables vanish again, but this time the statement is *true*.
Step 3. A true statement means every point of the line satisfies both equations: the two equations describe the same line, so there are infinitely many solutions — all pairs with $y = 3x - 4$. Taking $x = 2$ gives $y = 3(2) - 4 = 2$, one specific solution. $\boxed{(2,\ 2)}$

11. Rectangle: width x, length y, perimeter 46 m.
Step 1. "Length is 5 more than twice the width" gives $y = 2x + 5$; the perimeter gives $2y + 2x = 46$.
Step 2. The first is already isolated, so substitute: $2(2x+5)+2x = 46$, so $4x+10+2x = 46$ and $6x = 36$, giving $x = 6$.
Step 3. Back-substitute: $y = 2(6)+5 = 17$. Check the perimeter: $2(17)+2(6) = 34+12 = 46$ ✓
The rectangle is 6 m wide and 17 m long. $\boxed{(6,\ 17)}$

12. Mixture: x litres of 20% acid, y litres of 50% acid.
Step 1. The volumes must total 12 L, so the missing equation is $x+y = 12$; the acid balance is given as $0.2x + 0.5y = 3.6$.
Step 2. Isolate in the volume equation: $y = 12-x$, then substitute: $0.2x+0.5(12-x) = 3.6$.
Step 3. $0.2x + 6 - 0.5x = 3.6$, so $-0.3x = -2.4$ and $x = 8$; then $y = 12 - 8 = 4$. Check the acid: $0.2(8) + 0.5(4) = 1.6 + 2 = 3.6$ ✓
The chemist uses 8 L of the 20% solution and 4 L of the 50% solution. $\boxed{(8,\ 4)}$